Optimizing E-commerce Flash Sale Strategy:
A Data-Driven Approach to Product Selection and Pricing.

Executive Summary

This project utilizes SQL Server to analyze a large-scale e-commerce dataset (sourced via Kaggle) containing 10,000+ global transactions. The analysis focuses on supporting Flash Sale product selection by identifying high-traffic "Anchor" products and "Liquidation" items to optimize inventory turnover. The final output provides a practical framework for data-driven promotional strategies, balancing aggressive discounting with overall profitability.

1. Objectives

The objective of this project was to simulate a retail data environment to answer key business questions regarding promotional efficiency:

- **Data Preparation:** Migrate and clean raw CSV data into a structured SQL database.
- **Performance Analysis:** Calculate Profit Margins and Sales Velocity to assess product health.
- **Product Categorization:** Classify inventory into strategic groups (Stars, Potential, Liquidation) for targeted Flash Sale campaigns.

2. Data Preparation & Structure

2.1. Data Ingestion & Quality Control

The migration from raw CSV to SQL Server followed strict data integrity rules to ensure the dataset was "analysis-ready" upon creation:

- **Pre-defined Schema:** Unlike standard imports, table columns were explicitly defined with optimized data types during the CREATE TABLE phase to eliminate the need for mid-process conversions.
- **Precision Enforcement:** Critical financial metrics (Sales, Profit, UnitPrice) were assigned as DECIMAL(18, 2) to guarantee absolute mathematical precision for margin calculations.
- **Structural Integrity:** OrderID was established as the Primary Key to prevent duplication, while OrderDate was set to the DATE type to support time-series velocity analysis.

2.2. Data Dictionary

The database utilizes a denormalized flat-table structure (SalesData) to maximize query speed for complex aggregations:

- Table: SalesData
    - **Quantitative Metrics:** Sales, Profit, Quantity, UnitPrice, Discount (Numerical types; core inputs for Profitability & Velocity matrices).
    - **Categorical Attributes:** ProductName, Category, Sub-Category (Used for strategic product segmentation).

- **Operational Context:** OrderDate, PaymentMode, Region (Attributes for behavioral and logistical trend analysis).

3. Business Questions & SQL Analysis

In order to optimize the Flash Sale campaign, a series of strategic questions were formulated to categorize products based on profitability and sales performance.

List of Business Questions:

1. Which products should be prioritized for Flash Sale to maximize revenue, sales volume, and purchase frequency?
2. What discount range drives the highest increase in sales volume without significantly reducing profit?
3. Which underperforming products (low sales) show potential for growth through promotional campaigns?
4. Which product categories perform best under discounted conditions?

Question 1: *Which products should be prioritized for Flash Sale to maximize revenue, sales volume, and purchase frequency?*

```sql
SELECT
    ProductName,
    Category,
    -- Core metrics
    SUM(Quantity) AS Total_Quantity,
    SUM(Sales) AS Total_Revenue,
    SUM(Profit) AS Total_Profit,
    COUNT(DISTINCT OrderID) AS Order_Frequency,
    AVG(Discount) AS Avg_Discount
FROM SalesData
WHERE Sales > 0
GROUP BY ProductName, Category
```

```sql
Segmentation AS (
    SELECT *,
        CASE
            WHEN Revenue_Rank <= 10 AND Quantity_Rank <= 10 THEN 'Star Product'
            WHEN Revenue_Rank <= 10 THEN 'High Value Product'
            WHEN Quantity_Rank <= 10 OR Frequency_Rank <= 10 THEN 'High Demand Product'
            ELSE 'Others'
        END AS Product_Type
    FROM Scoring
)
```

(Full SQL script available on [GitHub](#))

	ProductName	Category	Total_Revenue	Total_Quantity	Order_Frequency	Total_Profit	Product_Type	Combined_Score
1	Headphones Accusantium	Electronics	857184.20	14	4	141873.79	Star Product	3.0
2	Lamp Enim	Home Decor	665920.20	16	4	93638.94	Star Product	4.0
3	Lamp Nostrum	Home Decor	596736.00	15	4	103319.16	High Demand Product	7.8
4	Textbook Omnis	Books	633246.25	14	3	119166.66	Star Product	10.9
5	Comics Voluptatibus	Books	571993.45	14	4	40250.95	High Demand Product	11.5
6	Puzzle Illo	Toys	593681.05	14	3	96106.40	High Demand Product	13.4
7	Oil Iusto	Groceries	544165.10	14	3	90875.36	High Demand Product	17.4
8	Wall Art Saepe	Home Decor	509854.10	14	4	109548.98	High Demand Product	21.0
9	Chair Adipisci	Furniture	512548.40	13	5	54777.81	High Demand Product	21.9
10	Doll Ab	Toys	487607.55	12	4	69246.23	High Demand Product	32.7

Insight:

The analysis reveals two key product groups suitable for Flash Sale campaigns: “Star Products” and “High Demand Products.”

Star Products generate the highest revenue while maintaining consistent purchase frequency, making them strong candidates for attracting traffic. Meanwhile, High Demand Products demonstrate stable sales volume and frequent transactions, indicating broad consumer appeal.

Additionally, the relatively consistent purchase frequency across products suggests that demand is distributed across multiple customers rather than driven by bulk purchases, reinforcing their suitability for mass-market promotional campaigns.

Recommendation: Flash Sale campaigns should prioritize a combination of Star Products and High Demand Products. Star Products can serve as anchor items to attract user traffic, while High Demand Products can drive transaction volume and engagement. A balanced portfolio approach will help maximize both GMV and platform activity, rather than relying on a small number of top-performing items.

Question 2: *What discount level drives the highest increase in sales volume without significantly reducing profit?*

```sql
CREATE OR ALTER VIEW Discount_View AS
SELECT
    Discount,
    SUM(Quantity) AS Total_Quantity,
    SUM(Profit) AS Total_Profit,
    SUM(Sales) AS Total_Revenue,
    SUM(Profit) * 1.0 / NULLIF(SUM(Sales),0) AS Profit_Margin
FROM SalesData
GROUP BY Discount;
GO

-- Query lại
SELECT *
FROM Discount_View
ORDER BY Discount;
```

(Full SQL script available on [GitHub])

	Discount	Total_Quantity	Total_Profit	Total_Revenue	Profit_Margin
1	0.00	2930	17266608.70	117106034.00	0.147444
2	5.00	3016	17059182.93	112096122.10	0.152183
3	10.00	3100	16562542.62	110572965.90	0.149788
4	15.00	2885	14575951.61	97285450.35	0.149826
5	20.00	3032	14244449.05	96605452.00	0.147449

Insight:
Data reveals that the 0% discount level yields the highest Total Revenue ($117M) and Profit ($17.2M). This indicates strong organic demand and high price inelasticity for the current product mix.

Increasing discounts to 10% successfully drives the highest traffic (3,100 units), yet fails to surpass the revenue or profit of the non-discounted state.

At 15% discount, we observe a sharp "performance valley" where volume drops despite deeper price cuts, signaling that excessive discounting may hurt perceived product value.

Recommendations:

Do not exceed a 10% discount threshold for Flash Sale events. Higher discounts significantly erode margins without providing a proportional boost in volume.

Use a 5-10% discount only for "Traffic Magnet" products to drive platform engagement, while maintaining 0% discount (original price) for high-demand "Star" products to protect overall bottom-line profitability.

Question 3: *Which underperforming products (low sales) show potential for growth through promotional campaigns?*

```sql
WITH ProductMetrics AS (
    SELECT
        ProductName,
        Category,
        SUM(Quantity) AS Total_Quantity,
        SUM(Profit) * 1.0 / NULLIF(SUM(Sales), 0) AS Margin,
        AVG(SUM(Quantity)) OVER() AS Avg_Quantity_System -- Lấy trung bình bán ra của cả kho
    FROM SalesData
    GROUP BY ProductName, Category
)
SELECT TOP 10
    ProductName,
    Category,
    Total_Quantity,
    CAST(Margin AS DECIMAL(18,2)) AS Profit_Margin
FROM ProductMetrics
WHERE Total_Quantity < Avg_Quantity_System -- Bán chậm hơn trung bình
  AND Margin > 0.20              -- Nhưng lợi nhuận biên vẫn rất dày (>20%)
ORDER BY Margin DESC;
```

(Full SQL script available on [GitHub])

	ProductName	Category	Total_Quantity	Profit_Margin
1	Table Deleniti	Furniture	1	0.25
2	Shampoo Hic	Beauty	1	0.25
3	Biography Repudiandae	Books	1	0.25
4	Sugar Ab	Groceries	1	0.25
5	Sofa Laudantium	Furniture	2	0.25
6	Spices Ab	Groceries	2	0.25
7	Table Nulla	Furniture	1	0.25
8	Rice Quos	Groceries	1	0.25
9	Women's Wear Et	Clothing	2	0.25
10	Cushion Ratione	Home Decor	1	0.25

Insight: "We identified a specific group of products that consistently maintains a high Profit Margin of 25%, significantly above the average but their sales velocity is critically low, with only 1-2 units sold over the period. These products are currently 'high-value, low-visibility'."

Recommendations:

Priority should be given to these "Hidden Gems" in upcoming Flash Sale campaigns. Even a 15% discount would leave a 10% net margin, which is still profitable while providing the price-cut necessary to trigger "impulse buying."

The primary goal for these products in a Flash Sale is not immediate profit maximization, but buying sales volume. Increasing the "Sold Count" from 1 to 50+ through a promotion will permanently improve their organic ranking (SEO) for long-term growth.

Question 4: *Which product categories perform best under discounted conditions?*

```sql
WITH CategoryPerformance AS (
    SELECT
        Category,
        -- Tổng số lượng bán ra khi KHÔNG có discount
        SUM(CASE WHEN Discount = 0 THEN Quantity ELSE 0 END) AS Total_Qty_Normal,
        -- Tổng số lượng bán ra khi CÓ discount > 0
        SUM(CASE WHEN Discount > 0 THEN Quantity ELSE 0 END) AS Total_Qty_Promo,
        -- Đếm số lượng giao dịch tương ứng để làm rõ quy mô
        COUNT(CASE WHEN Discount = 0 THEN 1 END) AS Transactions_Normal,
        COUNT(CASE WHEN Discount > 0 THEN 1 END) AS Transactions_Promo
    FROM SalesData
    GROUP BY Category
)
SELECT
    Category,
    Total_Qty_Normal,
    Total_Qty_Promo,
    -- Tính % tăng trưởng tổng sản lượng
    CAST(((Total_Qty_Promo - Total_Qty_Normal) * 100.0 / NULLIF(Total_Qty_Normal, 0)) AS DECIMAL(10,2)) AS Volume_Growth_Percent,
    -- Hiệu suất trung bình trên mỗi giao dịch khuyến mãi (để check lại)
    CAST((Total_Qty_Promo * 1.0 / NULLIF(Transactions_Promo, 0)) AS DECIMAL(10,2)) AS Items_Per_Promo_Order
FROM CategoryPerformance
ORDER BY Volume_Growth_Percent DESC;
```

(Full SQL script available on [GitHub](#))

	Category	Total_Qty_Normal	Total_Qty_Promo	Volume_Growth_Percent	Items_Per_Promo_Order
1	Kitchen	260	1284	393.85	2.92
2	Toys	251	1224	387.65	3.08
3	Clothing	275	1238	350.18	3.03
4	Home Decor	286	1253	338.11	3.00
5	Furniture	304	1287	323.36	3.01
6	Beauty	278	1117	301.80	3.01
7	Electronics	288	1138	295.14	3.10
8	Sports	302	1187	293.05	2.98
9	Groceries	302	1118	270.20	3.03
10	Books	384	1187	209.11	2.92

Insight:
The Kitchen and Toys categories are the most "Promotion-Responsive," exhibiting a massive volume surge of 394% and 388% respectively when discounted.

While total volume spiked significantly, the Items per Order remained stable (~3.0). This proves that promotions are highly effective at attracting a larger volume of customers rather than just increasing the number of items per existing customer.

All categories showed at least a 200% volume growth during promotional periods, indicating that discounts are a primary driver for market share expansion across the entire product portfolio.

Recommendations:
Operations should prepare inventory levels up to 4x higher than the daily baseline for the Kitchen category specifically to prevent out-of-stock (OOS) losses during sales peaks.

Upsell Strategy: Since the "items per order" is stagnant at 3.0, implement "Threshold Discounts" (e.g., Buy 4 units to get an extra 5% off) to push the average basket size higher for these high-growth categories.

4. Conclusion

The data-driven analysis of the global transaction dataset confirms that a successful Flash Sale strategy is not merely about "slashing prices," but about balancing volume growth with brand equity and profitability.

Core Findings Summary:
A 10% discount is the "sweet spot" for market engagement. Beyond this threshold, the brand risks a "Performance Valley" where perceived value drops and margins erode without a corresponding volume lift.

The Kitchen and Toys categories act as the primary engines for platform traffic, capable of achieving ~400% volume growth. These should lead every promotional campaign to maximize customer acquisition.

Products with 25% margins but low sales volume represent a significant untapped opportunity. Flash Sales should be used as a strategic tool to "buy" SEO ranking for these items through increased sales counts.